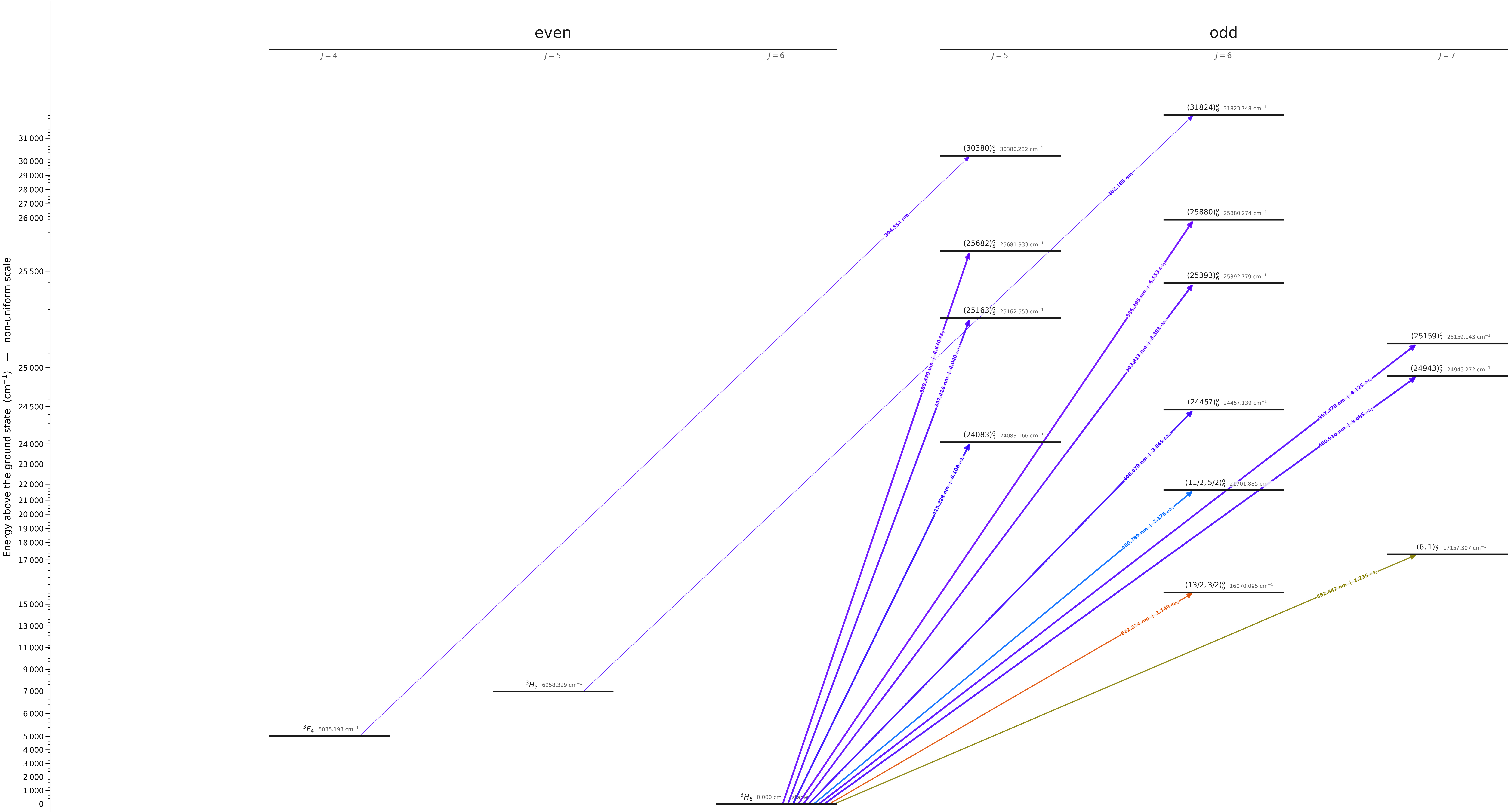
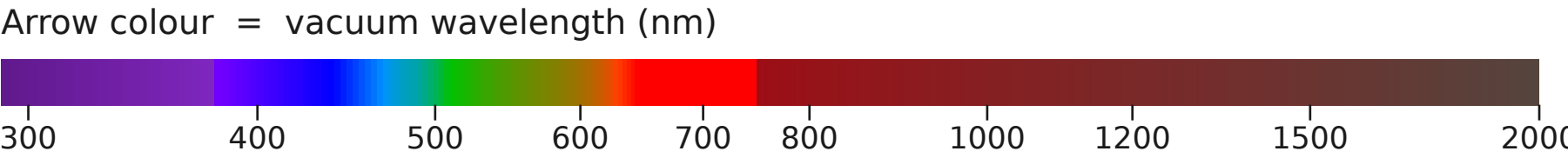


Erbium

Er (Er I). Strongest classified transitions below 2 μm (NIST): 16 levels, 13 lines.

Er⁺ ionisation limit 49262.00 cm⁻¹ = 6.10771 eV (202.996 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| er \| J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \ er \ J' \rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
³ H ₆ - (25880) ₆ ^o	386.3947	386.2851	775.8711	6.553	1.160e+08	-	-	-
³ H ₆ - (25682) ₅ ^o	389.3788	389.2685	769.9250	4.830	7.280e+07	-	-	-
³ H ₆ - (25393) ₆ ^o	393.8127	393.7013	761.2564	3.383	2.920e+07	-	-	-
³ H ₆ - (25163) ₅ ^o	397.4160	397.3036	754.3544	4.040	4.790e+07	-	-	-
³ H ₆ - (25159) ₅ ^o	397.4698	397.3574	754.2521	4.125	3.660e+07	-	-	-
³ H ₆ - (24943) ₅ ^o	400.9097	400.7964	747.7805	9.085	1.730e+08	-	-	-
³ H ₆ - (24457) ₆ ^o	408.8786	408.7632	733.2066	3.645	3.030e+07	-	-	-
³ H ₆ - (24083) ₅ ^o	415.2278	415.1107	721.9951	6.108	9.600e+07	-	-	-
³ H ₆ - (11/2, 5/2) ₆ ^o	460.7895	460.6604	650.6061	2.176	7.540e+06	-	-	-
³ H ₆ - (6, 1) ₅ ^o	582.8421	582.6805	514.3631	1.235	1.040e+06	-	-	-
³ H ₆ - (13/2, 3/2) ₆ ^o	622.2739	622.1017	481.7693	1.140	8.400e+05	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).